

Models and algorithms for Wireless Power Transfer in Ad-Hoc Networks and the IoT

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**MSc Web Intelligent Systems
Internet of Things**



- A **Wireless Power Transfer system** consists of:
 - **chargers** which transmit power wirelessly
 - **receivers** which harvest the energy from the chargers
- Envisioned by **Tesla**, is now becoming a game-changing technology¹
- **More reliable and controllable** than ambient energy harvesting (solar, light, wind, etc.)



¹ A. Kurs, A. Karalis, R. Moffatt, J. D. Joannopoulos, P. Fisher and M. Soljacic, Wireless power transfer via strongly coupled magnetic resonances, Science, vol. 317, pp. 83, 2007.

Two Enabling Technologies

- **Radio frequency(RF)**: low charging efficiency, only support low-power devices, charging distance up to few meters (0,13% efficiency).
- **Resonant magnetic coupling**: high charging efficiency, supports high-power equipment (e.g. electrical vehicles), charging distance ≤ 1 m (91 - 93% efficiency).



Figure: Electromagnetic radiation products by Powercast Corp.



Figure: Resonant magnetic coupling by WiTricity for electrical vehicle charging



- WPT has evolved into:
 - a subject of rapid **technological** progress
 - a very active **research** topic
 - a domain of emerging **practical development** and commercial application
- **Specialized** journals, conferences and first books have appeared.
- **Wireless Power Consortium** - Cooperation of Asian, European, and American companies in diverse industries. Working towards the **global standardization** charging technology (**Qi standard**).
- **Alliance for Wireless Power** - Independently operated organization composed of global wireless power and technology industry leaders (**Rezense standard**).
- **Commercial products** utilizing wireless power transfer are already available in the market.



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Wireless charging: How the key iPhone X and iPhone 8 feature will work



iPhone X and Apple Watch on the AirPower charging pad. © Apple

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Would YOU let a friend share your phone battery? Sony patents technology for wireless power transfer between devices

- Sony has patented tech to let you 'steal' battery power from nearby devices
- Phones could search for power in a similar way to seeking out Wi-Fi hot spots
- A chip with data transfer technology could enable power exchange wirelessly

By [DAISY DUNNE FOR MAILONLINE](#)

PUBLISHED: 12:47 BST, 17 March 2017 | **UPDATED:** 13:51 BST, 17 March 2017

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Wireless Power Market Surges Forward in 2017

April 20, 2017 | IHS Markit

🕒 Reading time 2 min (612 words)

IHS Markit today announced that the market for wireless power and charging is expected to grow in 2017, with global receiver shipments reaching 325 million units by year end. The market grew approximately 40% in 2016.

According to the new IHS Markit 2017 Wireless Power Tracker, mobile phones remain the primary driver for development of wireless power technology and mobile phones, laptops and wearable devices will drive receiver market growth. Receiver unit shipments are expected to climb this year as demand increases; OEMs make their decision on specifications; and the market moves towards glass-backed phones.

In 2016, the flagship Samsung handset made the biggest single contribution to enabled receiver shipments. Samsung's decision to retain wireless charging on the Galaxy S7, and LG's decision to add the technology to its new G6 device (U.S. model), were crucial for wireless charging technology. Apple is expected to follow suit later this year with its next iPhone, further boosting receiver shipments.

"The mobile phone market continues to be the entry point for the consumer experience of wireless power," said Vicky Yussuff, Analyst for IHS Markit. The IHS Markit Q1 consumer survey shows that consumer awareness of wireless power technology remains high and, perhaps more significantly, there has been a jump in usage. "More than one-third (35%) of U.S. consumers say they have used wireless charging, up from 25% in 2016. This suggests the market has moved into the commoditization phase, particularly for mobile phones." Yussuff said.

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7 APRIL 2017 BY MACIEJ HEYMAN

Wireless Charging Market Products Will Net Revenues Worth US\$ 27 Billion by 2024

New York, NY — ([SBWIRE](#)) — 04/07/2017 — As we head towards an era where wires won't necessarily be a requisite for powering up devices, tech companies from around the world are intensifying their efforts towards production of advanced and immaculate wireless charging systems. Innovative technologies supporting the working mechanism of wireless charging devices is also piquing the curiosity of tech enthusiasts as well as common consumers. From green vehicles to smartphones, Persistence Market Research has published its research study on how wireless charging is spanning the globe as the next-gen mode of charging electric devices.

Models and algorithms for Wireless Power Transfer in Ad-Hoc Networks and the IoT

10 / 50

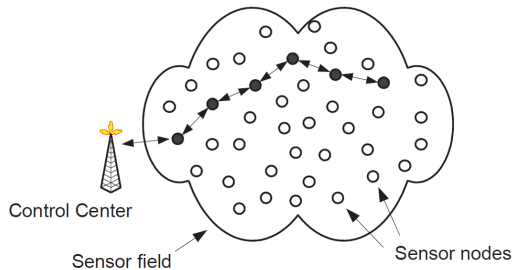


Scope of this talk

- Via the prism of Wireless Sensor Networks, we will initially discuss the **paradigm shift** that WPT represents in ad-hoc networks from an **algorithmic perspective**.
- We will discuss heuristics that high-light the underlying issues in static and dynamic settings.
- We will review recent results in terms of
 - **formal charging models**
 - WPT specific **concepts and phenomena**
 - key **optimization problems**



Wireless Sensor Networks (WSN)



The paradigm:

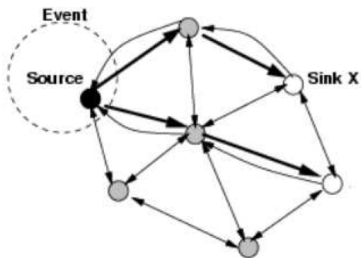
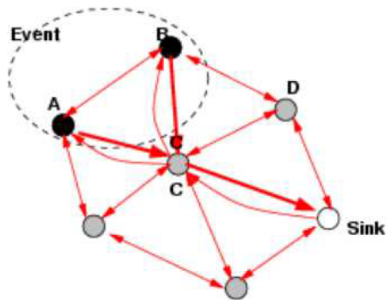
- Tiny, highly constrained autonomous devices with wireless capabilities
- Self-organized into an ad-hoc network to carry out a sensing task



Flat routing

All nodes
in the network have **similar** role regarding the routing of data. **No special nodes** are used.

Example: Directed Diffusion²

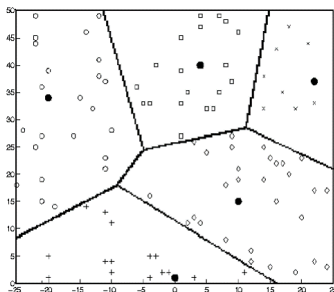
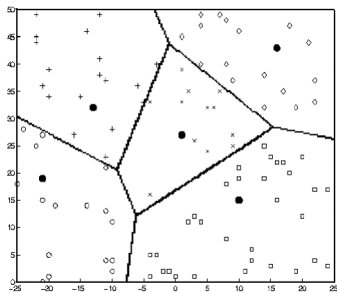


²Chalermek Intanagonwiwat, Ramesh Govindan, Deborah Estrin: Directed diffusion: a scalable and robust communication paradigm for sensor networks. MOBICOM 2000: 56-67

Hierarchical routing

Special nodes assume greater responsibility regarding routing of data than most nodes inside the network. Super nodes – **cluster heads**.

Example: LEACH³



³ - Wendi Rabiner Heinzelman, Anantha Chandrakasan, Hari Balakrishnan: Energy-Efficient Communication Protocol for Wireless Microsensor Networks. HICSS 2000

Energy is the Big Issue

Flat and Hierarchical routing are two fundamental routing paradigms. Other paradigms also exist (e.g. for obstacle avoidance) but energy is **The Issue**.

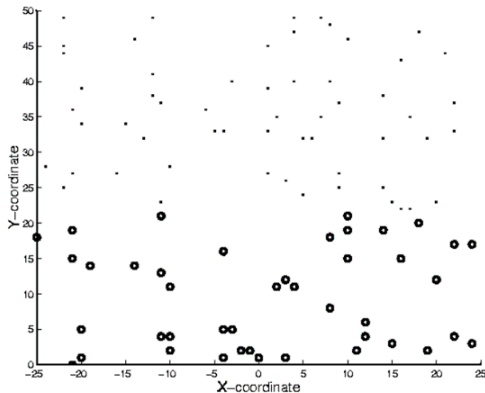


Figure: Multi-hop Transmissions

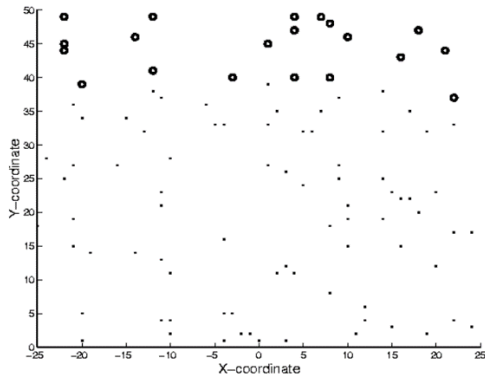


Figure: Direct Transmissions



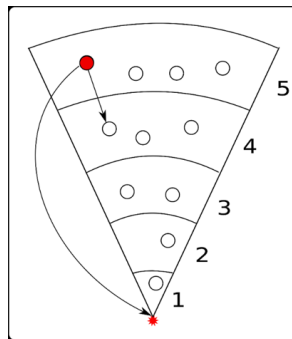
Network evolution over time. Sink in the far North.

The Energy Balance Protocol

Intuition: To avoid premature network disconnection, all nodes should dissipate energy **at the same rate independently of their position**⁴.

The protocol combines two transmission types:

- Short transmissions to the next sector with probability P_i
- Long direct transmissions to the Sink with probability $1 - P_i$
- The closer to the Sink, the more likely to send directly to the Sink



It is proven that EBP **maximises** the flow and the lifetime of the network. **The best we can do in terms of algorithms!**



⁴Charilaos Efthymiou, Sotiris E. Nikolettseas, José D. P. Rolim: Energy Balanced Data Propagation in Wireless Sensor Networks. IPDPS 2004

Constantinos Marios Angelopoulos, Sotiris E. Nikolettseas, Theofanis P. Raptis:
Wireless energy transfer in sensor networks with adaptive, limited knowledge protocols. *Computer Networks* 70: 113-141 (2014)



A new network paradigm

A **Wireless Rechargeable Sensor Network** consists of

- a **set of sensor motes:** that are deployed over an area of interest; the *network area*
- a **Sink:** a special node of the network towards which sensory data are routed
- a **Mobile Charger:** a *mobile* entity, with significant energy reserves, that can *recharge* the sensor motes via *wireless energy transfer*



Wireless Rechargeable Sensor Networks (WRSNs) enable:

- the highly constrained resource of **energy** to be managed in **great detail** and **more efficiently**
- the energy management to be performed **passively** by sensors, with no computational and communication overhead
- the energy management to be studied and designed **independently** of the underlying routing protocol



Remarks

- The wireless recharge problem in WSNs may look similar to other problems (e.g. data collection via mobile sinks)
- However, it admits **special features** and new **trade-offs** that necessitate a **direct approach**
- For instance, what is the amount of energy each mote should receive?
- Note that charger optimisation problems are (inherently) computationally hard (see NP-completeness of the *Charger Dispatch Decision Problem -CDDP*)⁵



⁵Constantinos Marios Angelopoulos, Sotiris E. Nikolettseas, Theofanis P. Raptis, Christoforos Raptopoulos, Filippos Vasilakis: **Efficient energy management in wireless rechargeable sensor networks**. MSWiM 2012: 309-316

We identify and try to optimize the following trade-offs:

- How should the total available *energy* of the network *be split between sensors and the Mobile Charger*?
- Given that the energy the Mobile Charger may deliver to the motes is finite, should each sensor be *fully or partially charged*?
- What **trajectory** should the MC follow in order to charge the sensors?

We seek to achieve **significant performance gains** over the no-recharge case.



The Charger Dispatch Decision Problem

Non-formal description: Given a set of sensors S , deployed over an area A , that generate and propagate data to a Sink, and a Mobile Charger M ;

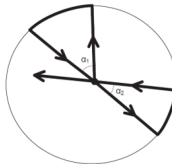
The *CDDP* is **to determine whether there is a feasible schedule for M** to visit the sensors so that no message is lost due to insufficient energy.

Theorem: CDDP is NP-complete. *Proof:* Reduction from Geometric-TSP.



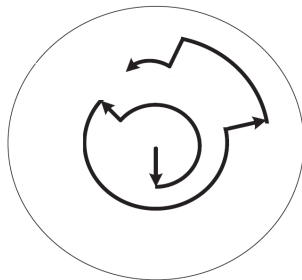
Heuristic Traversal Strategies

- **Global knowledge strategy:** The MC prioritizes motes based on distance and residual energy.
- **Random walk strategy:** The MC chooses the next mote u.a.r. among the neighbours of the current mote.
- **Space-filling strategy (the spiral):** Covers the entire network / avoids overlaps.
- **Move along the diameter strategy:** Quickly changes sub-regions.



The Adaptive Strategy

- The Charger follows **concentric** trajectories.
- Using **limited information, locally** available, it is able to identify areas of high energy depletion.
- The radius of the trajectory **adapts** to the energy depletion rates of each sub-region.

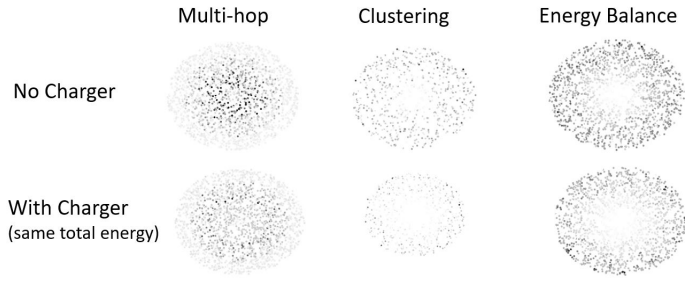


Criticality of node v_i at time t :

$$c_i(t) = f_i(t) \cdot \rho_i(t)$$

, where $f_i(t)$: **normalised traffic flow** at v_i and $\rho_i(t)$: **normalised energy consumption** at v_i . The protocol implicitly **adapts** to the underlying routing protocol.





- Significant performance gains achieved by introducing the charger.
- By using locally available network information the protocol achieves similar performance to the powerful global knowledge strategy.



Constantinos Marios Angelopoulos, Julia Buwaya, Orestis Evangelatos, Jose D. P. Rolim:
Traversal Strategies for Wireless Power Transfer in Mobile Ad-Hoc Networks.
*18th ACM International Conference on Modeling, Analysis and Simulation of Wireless and
Mobile Systems* , MSWiM 2015: 31-40



Definition

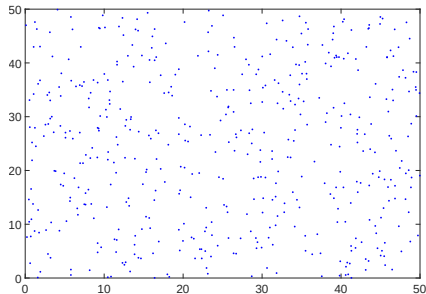
The Problem:

In an ad-hoc network where agents are randomly and dynamically moving inside an area of interest, what traversal strategy should a Mobile Charger (MC) follow in order to efficiently recharge them.

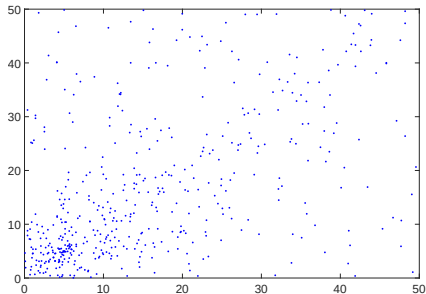


The Mobility Model

- The traversal speed of each agent is modelled as a Poisson random variable
- We consider two mobility models:
 - **Random Walk Mobility Model:** Each $\mathcal{A}_i$ performs an *independent blind random walk* $\rightarrow$ **uniform** agent distributions
 - **extended Random Walk Mobility Model:** Each $\mathcal{A}_i$ performs a *biased random walk* towards some *hotspots* $\rightarrow$ **heterogeneous** agent distributions (concept of **social attractivity**)



(a) Snapshot of a homogeneous agent distribution



(b) Snapshot of a heterogeneous agent distribution



- All agents are considered identical.
- The amount of dissipated energy for a given time interval is modelled as a Poisson random variable.

Charging Model:

- We consider that MC employs inductive charging and transmits energy omni-directionally → **one-to-many charging model**.
- For energy transmission we consider the Friis formula and the RF-to-DC energy conversion efficiency
- We assume that the energy reserves of MC are practically **inexhaustible**



The **Charger Traversal Decision Problem** (CTDP) is to determine whether there is a feasible schedule for MC to visit points in Ω such that no agent "dies" at any time.

Theorem: The CTDP is $\mathcal{NP}$ -complete.

Proof outline: Reduction from the Charger Dispatch Decision Problem - CDDP⁶
(reduced from the Geometric Travelling Salesman Problem)



⁶ Efficient energy management in wireless rechargeable sensor networks Constantinos Marios Angelopoulos, Sotiris E. Nikolettseas, Theofanis P. Raptis, Christoforos Raptopoulos, Filippos Vasilakis 15th ACM International Conference on Modeling, Analysis and Simulation of Wireless and Mobile Systems, Paphos, Cyprus, (MSWiM 2012).

The Weighting Function

The MC needs a way to prioritize the agents for charging. We identify the following network aspects:

- the **energy need** $E_i^{need} := E_{max} - E_i$ (with E_i indicating residual energy) of each agent A_i
- the mean energy **dissipation** rate λ_i of each agent A_i
- the **mobility** level $\mathcal{M}_{x(i)}^i$ of each agent A_i
- the euclidean **distance** d_i of each agent A_i from the MC



The Weighting Function

Definition (The Weighting Function)

$$W : \mathcal{A} \rightarrow \mathbb{R}_0^+; \quad W_i \mapsto \frac{(\bar{E}_i^{need})^\alpha (\bar{\lambda}_i)^\beta}{(\bar{\mathcal{M}}_{x(i)}^i)^\gamma (\bar{d}_i)^\delta} \quad (1)$$

where $\bar{E}_i^{need} := \frac{E_{max} - E_i}{E_{max}} \cdot 100$; $\bar{\lambda}_i := \frac{\lambda_i}{\lambda_{max}} \cdot 100$; $\bar{\mathcal{M}}_{x(i)}^i := \frac{\mathcal{M}_{x(i)}^i}{\mathcal{M}_{max}} \cdot 100$; $\bar{d}_i := \frac{d_i}{d_{max}} \cdot 100$

- We use normalized values due to different units and scales of each parameter
- Nominator and denominator are defined based on the desired monotonicity of W
- Exponents are positive integers that help fine tune the importance of each parameter



Definition (The Global Knowledge ILP)

$$\max_{x,y,\beta_i,d_i} \sum_i W_i \beta_i \quad (2a)$$

$$d_i \geq |x_i - x| \quad \forall i \quad (2b)$$

$$d_i \geq |y_i - y| \quad \forall i \quad (2c)$$

$$R_{low} + (1 - \beta_i) \cdot \mathbf{M} \geq d_i \quad \forall i \quad (2d)$$

$$\beta_i \in \{0, 1\} \quad \forall i \quad (2e)$$

i : the indice of agent $A_i \in \mathcal{A}$; W_i : the weight of $A_i \in \mathcal{A}$

β_i : binary decision variable

d_i : the distance of A_i from the MC (x, y)

x_i, y_i : the position of A_i and $\mathbf{M}$ a large constant



- **The Global Knowledge Tessellation Strategy:** Area Ω is virtually tessellated into tiles; the Charger optimizes its trajectory according to the cumulative weight of each tile.
- **The Space-filling Strategy:** The Charger systematically sweeps Ω in a deterministic way. Requires **some network knowledge**
- **The Random Walk Strategy:** The Charger performs a random walk in Ω . **Zero-knowledge, randomized** traversal strategy.



Local knowledge traversal strategy that exploits **local information** collected **distributedly**.

Agents assume an active role by collecting local network information periodically:

- Current coordinates (x_i, y_i)
- Total energy needed by one-hop neighbours $(E_i^{\bar{need}})$
- Average energy dissipation rate of neighbourhood $(\bar{\lambda}_i)$
- Average mobility level of neighbourhood $(\mathcal{M}_{x(i)}^{\bar{i}})$



The Reactive Local Knowledge Strategy -RLK

From this collected information each agent computes a weight via a slightly modified weighting function (distance to MC is missing)

$$\bar{W} : \bar{W}_i \mapsto \frac{E_i^{\bar{need}} \bar{\lambda}_i}{\mathcal{M}_{x(i)}^{\bar{i}}} \quad (3)$$

This info along with the local coordinates are stored and carried during the agent's network traversal in the form of tuples.

Tuples *age* over time and are *updated* with *information of higher quality*, when found.

The MC gains a “blurred” view of the network status in the near past and adjusts its traversal



Some remarks:

- The protocol introduces some communication overhead but this is limited and local.
- Information regarding critical areas are carried for more time and thus further in the network.
- The protocol is distributed and therefore scalable and applicable in realistic settings.

Performance Evaluation:

- In homogeneous settings cheap naive strategies, such as Random Walks, demonstrate good performance
- In heterogeneous settings introducing a small network overhead in a sophisticated way drastically increases performance



S. Nikolettseas, T. Raptis, C. Raptopoulos. **Low Radiation Efficient Wireless Energy Transfer in Wireless Distributed Systems**, in ICDCS 2015.

Also in the Journal of Computer Networks (COMNET 2017).



The Scalar Charging Model

- Authors define a **novel charging model** that takes into account *real hardware* restrictions:
 - **finite initial charger energy supplies** (it can transfer restricted energy)
 - **finite node battery capacity** (it can store restricted energy)
 - ⇒ non-linear constraints (time multiplies power) that radically change the complexity of the considered computational problems
 - ⇒ authors focus on **useful energy** received (not just power rate)

Charging: A node $v \in P$ harvests energy from a charger $u \in M$ with charging rate given by

$$P_{v,u}(t) = \begin{cases} \frac{\alpha r_u^2}{(\beta + \text{dist}(v,u))^2}, & \text{if } E_u^{(t)}, C_v^{(t)} > 0, \text{dist}(v,u) \leq r_u \\ 0, & \text{otherwise} \end{cases}$$

α and β are known positive constants determined by the environment and hardware.



Energy: The harvested energy by the nodes is **additive**. Therefore, the total energy that node v gets within the time interval $[0, T]$ is

$$H_v(T) = \sum_{u \in M} \int_0^T P_{v,u}(t) dt$$

EMR: The **electromagnetic radiation (EMR)** at a point x is proportional to the additive power received at that point. In particular, for any $x \in A$, the EMR at time t on x is given by

$$R_x(t) = \gamma \sum_{u \in M} P_{x,u}(T)$$

where γ is a constant that depends on the environment.



The Low Radiation Efficient Charging (LREC) Problem

Authors define the following:

- Let M a set of **chargers** and P be a set of **nodes** in an area A . Suppose that
 - (i) each charger $u \in M$ initially has available energy $E_u^{(0)}$
 - (ii) each node $v \in P$ has initial energy storage capacity $C_v^{(0)}$
- Assign to each charger $u \in M$ a radius r_u , so that
 - (a) the total **usable energy** given to the nodes of the network **is maximized**
 - (b) the electromagnetic radiation at any point of A is **at most** ρ .

The problem is proven to be **NP-hard** (the problem is relaxed and an ILP formulation is provided).

Authors, also provide interesting insights in fundamental properties of the problem; e.g. by introducing the power constraint, affects the monotonicity of the objective function.



I. Katsidimas, S. Nikoletseas, T. Raptis and C. Raptopoulos. **Efficient Algorithms for Power Maximization in the Vector Model for Wireless Energy Transfer**, in ICDCN 2017.

Also in the Journal of Pervasive and Mobile Computing (PMC), 2017.



- **Friis model** (widely used):

$$P_r = P_t \times G_t \times G_r \times \left(\frac{\lambda}{4\pi R} \right)^2$$

where G_t , G_r antenna gains, λ the wavelength and R the distance.

- It is a **scalar (1-dimensional) model**, assuming received energy is **additive**.
- A gross model, **unable to explain detailed phenomena** in real applications with many nodes.
- Still, in case of one charger or few remote chargers, it is valid. When "micro management" is needed in the presence of **several nearby chargers**, then it is **not sufficient**.



A need for more precise models

- **Multiple nearby transmitters** introduce **complex interference** among waves.
- This leads to interesting **constructive and destructive** combinations of waves.
- This complex process necessitates the introduction of **vector (2-dimensional)** models, providing us with **more detailed and realistic** modelling abstractions.



- The electric field created by an energy transmitter (charger) C , operating at full capacity, at a receiver R at distance $d = \text{dist}(C, R)$ is given by⁷

$$E(C, R) \stackrel{\text{def}}{=} \sqrt{\frac{Z_0 \cdot G_C \cdot P_C}{4 \cdot \pi d^2}} \cdot e^{-j \frac{2\pi}{\lambda} d}$$

where Z_0 a constant for wave-impedance, G_C the antenna gain, P_C the transmitted power and λ the wavelength.

- This gives rise to a **2-dimensional vector**:

$$E(C, R) \stackrel{\text{def}}{=} \beta \cdot \frac{1}{d} \cdot e^{-j \frac{2\pi}{\lambda} d} = \beta \cdot \frac{1}{d} \cdot \begin{bmatrix} \cos(\frac{2\pi}{\lambda} d) \\ \sin(\frac{2\pi}{\lambda} d) \end{bmatrix}$$

where β constant depends on hardware and environment.



⁷M. Y. Naderi, K. R. Chowdhury, S. Basagni, Wireless sensor networks with RF energy harvesting: Energy models and analysis, in WCNC 2015

- The total electric field at a receiver R created by a **family of energy transmitters** $\mathcal{C}$ is the **superposition (vector-sum)** of their individual electric fields, that is

$$E(\mathcal{C}, R) \stackrel{\text{def}}{=} \sum_{C \in \mathcal{C}} E(C, R)$$

.

- Furthermore, the **total available power** at the receiver R is given by

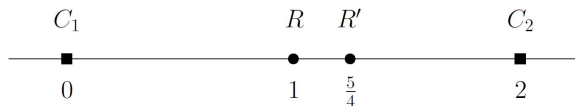
$$P(\mathcal{C}, R) = \gamma \cdot \|E(\mathcal{C}, R)\|^2$$

where $\|\cdot\|$ denotes the **length (2-norm)** of the vector and γ constant that depends on hardware.



Interesting Phenomena - how two nearby points can differ a lot

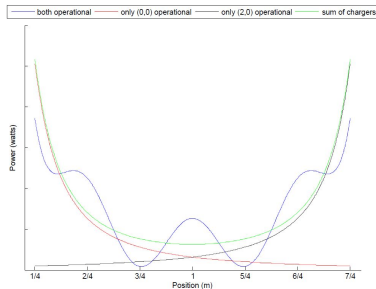
- charger
- node



- $P(C_1, R) = P(C_2, R) = 1$ (only 1 transmitter operational).
- $P(C_1, C_2, R) = (1 + 1)^2$ (both operational $\Rightarrow$ **superadditive** power since C_1, C_2 are equidistant and vectors have same direction).
- $P(C_1, C_2, R') = \frac{8}{15}^2 \sim 0.28 < \min P(C_1, R'), P(C_2, R') = \frac{4}{5}^2 \sim 0.64$ (**cancellation** effect at a nearby point).



Interesting Phenomena - the full curves



- **Local maxima** indicate points of **superadditive** power.
- **Local minima** occur at points of **cancellation**.
- **Note:** It is highly **non-trivial** to **derive a closed formula** for the points of maximum/minimum power. So we have to examine **a whole space of system configurations** and evaluate power received.



Two optimization problems

Definition (MAX-POWER)

Given a family of chargers $\mathcal{C}$ and family of receivers $\mathcal{R}$, **find a configuration for the chargers that maximizes the total power** to $\mathcal{R}$. That is, find x^* such that

$$x^* \in \arg \max_{x \in [0,1]^{\mathcal{C}}} P(\mathcal{C}(x), \mathcal{R})$$

where $P(\mathcal{C}(x), \mathcal{R}) = \sum_{R \in \mathcal{R}} P(\mathcal{C}(x), R)$

Definition (MAX- k MIN-GUARANTEE)

Given a family of chargers $\mathcal{C}$ and a family of receivers $\mathcal{R}$, **find a configuration for the chargers that maximizes the minimum cumulative power among all subsets of $\mathcal{R}$ of size k** . That is, find x^* such that

$$x^* \in \arg \max_{x \in [0,1]^{\mathcal{C}}} \min_{A \in \binom{\mathcal{R}}{k}} P(\mathcal{C}(x), A)$$

where $P(\mathcal{C}(x), A) = \sum_{R \in A} P(\mathcal{C}(x), R)$.



Until next time!

